Power Question 2014: Power of Potlucks

In each town in ARMLandia, the residents have formed groups, which meet each week to share math problems and enjoy each others' company over a potluck-style dinner. Each town resident belongs to exactly one group. Every week, each resident is required to make one dish and to bring it to his/her group.

It so happens that each resident knows how to make precisely two dishes. Moreover, no two residents of a town know how to make the same pair of dishes. Shown below are two example towns. In the left column are the names of the town's residents. Adjacent to each name is the list of dishes that the corresponding resident knows how to make.

ARMLton

Resident	Dishes
Paul	pie, turkey
Arnold	pie, salad
Kelly	salad, broth

ARMLville

Resident	Dishes
Sally	steak, calzones
Ross	calzones, pancakes
David	steak, pancakes

The *population* of a town T, denoted $\text{pop}(T)$, is the number of residents of T. Formally, the town itself is simply the set of its residents, denoted by $\{r_1, \ldots, r_{\text{pop}(T)}\}$ unless otherwise specified. The set of dishes that the residents of T collectively know how to make is denoted $\text{dish}(T)$. For example, in the town of ARMLton described above, $\text{pop}(\text{ARMLton}) = 3$, and $\text{dish}(\text{ARMLton}) = \{\text{pie, turkey, salad, broth}\}$.

A town T is called *full* if for every pair of dishes in $\text{dish}(T)$, there is exactly one resident in T who knows how to make those two dishes. In the examples above, ARMLville is a full town, but ARMLton is not, because (for example) nobody in ARMLton knows how to make both turkey and salad.

Denote by $\mathcal{F}_d$ a full town in which collectively the residents know how to make d dishes. That is, $|\text{dish}(\mathcal{F}_d)| = d$.

1a. Compute $\text{pop}(\mathcal{F}_{17})$. [1 pt]

1b. Let $n = \text{pop}(\mathcal{F}_d)$. In terms of n, compute d. [1 pt]

1c. Let T be a full town and let $D \in \text{dish}(T)$. Let T' be the town consisting of all residents of T who do not know how to make D. Prove that T' is full. [2 pts]

In order to avoid the embarrassing situation where two people bring the same dish to a group dinner, if two people know how to make a common dish, they are forbidden from participating in the same group meeting. Formally, a *group assignment* on T is a function $f : T \to \{1, 2, \ldots, k\}$, satisfying the condition that if $f(r_i) = f(r_j)$ for $i \neq j$, then r_i and r_j do not know any of the same recipes. The *group number* of a town T, denoted $\text{gr}(T)$, is the least positive integer k for which there exists a group assignment on T.

For example, consider once again the town of ARMLton. A valid group assignment would be $f(\text{Paul}) = f(\text{Kelly}) = 1$ and $f(\text{Arnold}) = 2$. The function which gives the value 1 to each resident

of ARMLton is **not** a group assignment, because Paul and Arnold must be assigned to different groups.

2a. Show that $\text{gr}(\text{ARMLton}) = 2$. [1 pt]

2b. Show that $\text{gr}(\text{ARMLville}) = 3$. [1 pt]

3a. Show that $\text{gr}(\mathcal{F}_4) = 3$. [1 pt]

3b. Show that $\text{gr}(\mathcal{F}_5) = 5$. [2 pts]

3c. Show that $\text{gr}(\mathcal{F}_6) = 5$. [2 pts]

4. Prove that the sequence $\text{gr}(\mathcal{F}_2), \text{gr}(\mathcal{F}_3), \text{gr}(\mathcal{F}_4), \ldots$ is a non-decreasing sequence. [2 pts]

For a dish D, a resident is called a *D-chef* if he or she knows how to make the dish D. Define $\text{chef}_T(D)$ to be the set of residents in T who are D-chefs. For example, in ARMLville, David is a steak-chef and a pancakes-chef. Further, $\text{chef}_{\text{ARMLville}}(\text{steak}) = \{\text{Sally, David}\}$.

5. Prove that
$$\sum_{D \in \text{dish}(T)} |\text{chef}_T(D)| = 2\,\text{pop}(T).$$ [2 pts]

6. Show that for any town T and any $D \in \text{dish}(T)$, $\text{gr}(T) \geq |\text{chef}_T(D)|$. [2 pts]

If $\text{gr}(T) = |\text{chef}_T(D)|$ for some $D \in \text{dish}(T)$, then T is called *homogeneous*. If $\text{gr}(T) > |\text{chef}_T(D)|$ for each dish $D \in \text{dish}(T)$, then T is called *heterogeneous*. For example, ARMLton is homogeneous, because $\text{gr}(\text{ARMLton}) = 2$ and exactly two chefs make pie, but ARMLville is heterogeneous, because even though each dish is only cooked by two chefs, $\text{gr}(\text{ARMLville}) = 3$.

7. For $n = 5, 6$, and 7, find a heterogeneous town T of population 5 for which $|\text{dish}(T)| = n$. [3 pts]

A *resident cycle* is a sequence of distinct residents $r_1, \ldots, r_n$ such that for each $1 \leq i \leq n-1$, the residents r_i and r_{i+1} know how to make a common dish, residents r_n and r_1 know how to make a common dish, and no other pair of residents r_i and r_j, $1 \leq i, j \leq n$ know how to make a common dish. Two resident cycles are *indistinguishable* if they contain the same residents (in any order), and distinguishable otherwise. For example, if r_1, r_2, r_3, r_4 is a resident cycle, then r_2, r_1, r_4, r_3 and r_3, r_2, r_1, r_4 are indistinguishable resident cycles.

8a. Compute the number of distinguishable resident cycles of length 6 in $\mathcal{F}_8$. [1 pt]

8b. In terms of k and d, find the number of distinguishable resident cycles of length k in $\mathcal{F}_d$. [1 pt]

9. Let T be a town with at least two residents that has a single resident cycle that contains every resident. Prove that T is homogeneous if and only if $\text{pop}(T)$ is even. [3 pts]

10. Let T be a town such that, for each $D \in \text{dish}(T)$, $|\text{chef}_T(D)| = 2$.

    a. Prove that there are finitely many resident cycles $C_1, C_2, \ldots, C_j$ in T so that each resident belongs to exactly one of the C_i. [3 pts]

    b. Prove that if $\text{pop}(T)$ is odd, then T is heterogeneous. [3 pts]

11. Let T be a town such that, for each $D \in \text{dish}(T)$, $|\text{chef}_T(D)| = 3$.

    a. Either find such a town T for which $|\text{dish}(T)|$ is odd, or show that no such town exists. [2 pts]

    b. Prove that if T contains a resident cycle such that for every dish $D \in \text{dish}(T)$, there exists a chef in the cycle that can prepare D, then $\text{gr}(T) = 3$. [3 pts]

12. Let k be a positive integer, and let T be a town in which $|\text{chef}_T(D)| = k$ for every dish $D \in \text{dish}(T)$. Suppose further that $|\text{dish}(T)|$ is odd.

    a. Show that k is even. [1 pt]

    b. Prove the following: for every group in T, there is some dish $D \in \text{dish}(T)$ such that no one in the group is a D-chef. [3 pts]

    c. Prove that $\text{gr}(T) > k$. [3 pts]

13a. For each odd positive integer $d \geq 3$, prove that $\text{gr}(\mathcal{F}_d) = d$. [3 pts]

13b. For each even positive integer d, prove that $\text{gr}(\mathcal{F}_d) = d - 1$. [4 pts]